Department of Chemical Engineering, Chulalongkorn University (teaching)

MEA CO2 Capture Ver 1 · operator training simulator

HEAT AND MATERIAL BALANCE, LINE SIZING AND PUMP HYDRAULICS

MEA CO2 capture unit: absorber T-101 / stripper T-102 (30 wt% MEA)

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Revision record

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Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

Teaching document for the MEA CO2 capture operator training simulator. Not for construction. Checker and approver are blank: this issue has not been checked or approved.

Contents

1 · Document, design basis and cases . . . 2
2 · Stream tables . . . 3
3 · Unit balances and duties . . . 9
4 · Off-design cases: what changes . . . 10
5 · Line list and line sizing . . . 11
6 · Pump and blower hydraulics . . . 12
7 · Design margins . . . 15
8 · Assumptions and holds (not design data) . . . 15
9 · Engineering queries . . . 16

1 · Document, design basis and cases

The heat and material balance of the MEA CO2 capture unit, written from the simulator model. Design case A is the page's design state (snapDesign: warm start 1200 s, noise off). Cases B and C are steady-state runs of the same model with the Operating-windows engine owRun: trips moved out of reach on the copy, the FIC-201 set point trimmed by a slow outer loop to hold 90 % capture, run until capture, lean loading, steam and TIC-206 settle. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

Case	Definition	Flue gas kNm³/h	CO2 mol%	Capture %	CO2 product t/h	Lean t/h	α lean / rich	Steam t/h	SRD GJ/t	Flood T-101 / T-102 %	Settled after
A Design case	176.8 kNm³/h flue gas, 4.0 mol% CO2, lean 269.7 t/h, TIC-206 120 °C (snapDesign, noise off)	176.8	4.00	90.03	12.50	269.7	0.246 / 0.471	22.94	3.94	62.1 / 50.6	design state
B Off-design 1 · 6 % CO2, lean raised	100 % flue gas at 6.0 mol% CO2 (top of the design range); FIC-201 raised by the outer capture loop to hold 90 % capture; TIC-206 120 °C	176.8	6.00	90.10	18.80	473.3	0.274 / 0.471	34.97	3.99	68.2 / 76.7	90 min
C Off-design 2 · 70 % turndown	70 % flue gas (123.8 kNm³/h) at 4.0 mol% CO2; FIC-201 trimmed to hold 90 % capture; TIC-206 120 °C	123.8	4.00	90.02	8.75	173.2	0.224 / 0.472	16.96	4.16	43.4 / 36.6	120 min

Design basis (model): NGCC flue gas after the DCC, 176.8 kNm³/h, 4.0 mol% CO2 (range 3–6 %), 40 °C saturated; 30 wt% MEA; 90 % capture, 300 t/d CO2; absorber T-101 D 5.4 m, 2 × 8.5 m Mellapak Plus 252Y + 3 m water wash; stripper T-102 D 2.6 m, 8 m IMTP 50, reboiler 120 °C, V-101 175 kPa a; LP steam 3.5 bar a and CW 30 °C assumed. Stream numbers are those of PFD MEA-PFD-001; line numbers those of P&ID CC-PID-101 (sizes TBC).

2 · Stream tables

Case A · Design case — streams 1 to 12

Stream no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
BFD no. (Process tab)		1				2	3	4		7	5	
Description	Flue gas from DCC	Flue gas to absorber	Wash water draw-off	Wash water to cooler	Cooled wash water	Treated gas to stack	Lean amine to absorber	Rich amine from sump	Rich amine pumped	Lean amine, warm	Rich amine, hot	Lean amine pumped
From	battery limit (DCC)	K-101	T-101 chimney tray	P-104	E-105	T-101	E-104 / FV-201	T-101	P-101 / FV-109	E-101	E-101	P-102
To	K-101	T-101	P-104	E-105	T-101 wash bed	stack	T-101	P-101	E-101	E-104	T-102	E-101
Phase	gas	gas	liquid	liquid	liquid	gas	liquid	liquid	liquid	liquid	liquid	liquid
Mass flow, t/h	**224.84**	**224.84**	**456.38**	**456.38**	**456.38**	**212.35**	**269.70**	**282.19**	**282.19**	**269.70**	**282.19**	**269.70**
Molar flow, kmol/h	7887.9	7887.9	—	—	—	7604.7	11517.0	11800.4	11800.4	11517.0	11800.4	11517.0
Temperature, °C	**40.0**	**40.0**	**47.9**	**47.9**	**37.0**	**40.0**	**40.0**	**43.6**	**43.6**	**51.6**	**109.0**	**120.0**
Pressure, kPa a	**102**	**107**	**104**	**468**	**417**	**103**	**403**	**122**	**504**	**453**	**295**	**495**
Pressure from	model	model	model	hydraulics	hydraulics	model	allowance	allowance	hydraulics	allowance	allowance	hydraulics
Density, kg/m³	1.120	1.171	989.0	989.0	993.4	1.105	1064.0	1112.1	1112.1	1057.7	1076.2	1020.0
Actual volume flow, m³/h	200747	191932	461	461	459	192224	253	254	254	255	262	264
Viscosity, mPa s	0.018	0.018	0.564	0.564	0.690	0.018	2.393	2.579	2.579	1.949	0.976	0.743
CO2, mol%	4.00	4.00				0.41						
H2O, mol%	6.94	6.94				7.21						
N2 + Ar, mol%	77.07	77.07				79.94						
O2, mol%	11.99	11.99				12.44						
Loading α, mol/mol							0.246	0.471	0.471	0.246	0.471	0.246
MEA, wt% (apparent)							28.50	27.23	27.23	28.50	27.23	28.50
CO2, wt% (apparent)							5.04	9.25	9.25	5.04	9.25	5.04
H2O, wt% (apparent)							66.46	63.52	63.52	66.46	63.52	66.46

Case A · Design case — streams 13 to LC

Stream no.	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**LS**	**LC**
BFD no. (Process tab)	8		10		9	12			6	11	
Description	Overhead vapour	Condenser outlet (two-phase)	CO2 product	Reflux water from drum	Reflux to stripper	Make-up water	Stripper bottoms to reboiler	Reboiler vapour	Lean amine from reboiler	LP steam	Steam condensate
From	T-102	E-103	V-101 / PV-301	V-101	P-103 / FV-303	demin header / FV-208	T-102 sump	E-102	E-102	LP steam header	E-102 (tubes)
To	E-103	V-101	compression	P-103	T-102	T-102 sump	E-102	T-102	P-102	E-102 (tubes)	condensate return
Phase	gas	two-phase	gas	liquid	liquid	liquid	liquid	gas	liquid	steam	liquid
Mass flow, t/h	**18.27**	**18.27**	**12.73**	**5.54**	**5.54**	**0.25**	**291.36**	**21.66**	**269.70**	**22.94**	**22.94**
Molar flow, kmol/h	604.2	604.3	296.6	307.6	307.6	13.7	12379.7	1106.8	11517.0	1273.4	1273.4
Temperature, °C	**103.1**	**40.0**	**40.0**	**40.0**	**40.0**	**35.0**	**118.0**	**120.0**	**120.0**	**138.9**	**138.9**
Pressure, kPa a	**181**	**175**	**175**	**175**	**325**	**400**	**182**	**182**	**182**	**350**	**350**
Pressure from	model	model	model	model	allowance	allowance	model	model	model	model	model
Density, kg/m³	1.748	4.135	2.884	992.2	992.2	994.1	1033.2	1.092	1020.0	1.841	926.1
Actual volume flow, m³/h	10451	4419	4413	5.59	5.59	0.25	282	19838	264	12464	25
Viscosity, mPa s	0.014	0.015	0.015	0.651	0.651	0.718	0.787	0.013	0.743	0.013	0.197
CO2, mol%	47.01	95.76	95.76					5.98		0.00	
H2O, mol%	52.99	4.24	4.24					94.02		100.00	
N2 + Ar, mol%	0.00	0.00	0.00					0.00		0.00	
O2, mol%	0.00	0.00	0.00					0.00		0.00	
Loading α, mol/mol							0.298		0.246		
MEA, wt% (apparent)							28.19		28.50		
CO2, wt% (apparent)							6.06		5.04		
H2O, wt% (apparent)							65.75		66.46		

Stream 11 is written as liquid as in the model: it flashes (EQ-P03). Stream 19 = 20 + 21 at steady state. Streams 3–5: assumed wash-water circulation. Pressure from: model = calculated by the model; allowance = the model's fixed allowance above column pressure; hydraulics = pump discharge (section 6).

Case B · Off-design 1 · 6 % CO2, lean raised — streams 1 to 12

Stream no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
BFD no. (Process tab)		1				2	3	4		7	5	
Description	Flue gas from DCC	Flue gas to absorber	Wash water draw-off	Wash water to cooler	Cooled wash water	Treated gas to stack	Lean amine to absorber	Rich amine from sump	Rich amine pumped	Lean amine, warm	Rich amine, hot	Lean amine pumped
From	battery limit (DCC)	K-101	T-101 chimney tray	P-104	E-105	T-101	E-104 / FV-201	T-101	P-101 / FV-109	E-101	E-101	P-102
To	K-101	T-101	P-104	E-105	T-101 wash bed	stack	T-101	P-101	E-101	E-104	T-102	E-101
Phase	gas	gas	liquid	liquid	liquid	gas	liquid	liquid	liquid	liquid	liquid	liquid
Mass flow, t/h	**227.28**	**227.28**	**456.38**	**456.38**	**456.38**	**208.37**	**473.33**	**492.83**	**492.83**	**473.33**	**492.83**	**473.33**
Molar flow, kmol/h	7888.0	7888.0	—	—	—	7453.2	20247.0	20704.7	20704.7	20247.1	20704.7	20247.1
Temperature, °C	**40.0**	**40.0**	**52.6**	**52.6**	**37.0**	**40.0**	**42.4**	**46.7**	**46.7**	**56.4**	**107.7**	**120.0**
Pressure, kPa a	**102**	**107**	**104**	**468**	**417**	**103**	**403**	**122**	**559**	**453**	**295**	**581**
Pressure from	model	model	model	hydraulics	hydraulics	model	allowance	allowance	hydraulics	allowance	allowance	hydraulics
Density, kg/m³	1.132	1.188	986.9	986.9	993.4	1.106	1066.5	1107.0	1107.0	1058.8	1073.4	1023.8
Actual volume flow, m³/h	200748	191353	462	462	459	188398	444	445	445	447	459	462
Viscosity, mPa s	0.018	0.018	0.521	0.521	0.690	0.018	2.335	2.443	2.443	1.831	0.991	0.757
CO2, mol%	6.00	6.00				0.63						
H2O, mol%	6.92	6.92				7.21						
N2 + Ar, mol%	75.36	75.36				79.75						
O2, mol%	11.72	11.72				12.41						
Loading α, mol/mol							0.274	0.471	0.471	0.274	0.471	0.274
MEA, wt% (apparent)							27.91	26.85	26.85	27.91	26.85	27.91
CO2, wt% (apparent)							5.52	9.12	9.12	5.52	9.12	5.52
H2O, wt% (apparent)							66.57	64.03	64.03	66.57	64.03	66.57

Case B · Off-design 1 · 6 % CO2, lean raised — streams 13 to LC

Stream no.	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**LS**	**LC**
BFD no. (Process tab)	8		10		9	12			6	11	
Description	Overhead vapour	Condenser outlet (two-phase)	CO2 product	Reflux water from drum	Reflux to stripper	Make-up water	Stripper bottoms to reboiler	Reboiler vapour	Lean amine from reboiler	LP steam	Steam condensate
From	T-102	E-103	V-101 / PV-301	V-101	P-103 / FV-303	demin header / FV-208	T-102 sump	E-102	E-102	LP steam header	E-102 (tubes)
To	E-103	V-101	compression	P-103	T-102	T-102 sump	E-102	T-102	P-102	E-102 (tubes)	condensate return
Phase	gas	two-phase	gas	liquid	liquid	liquid	liquid	gas	liquid	steam	liquid
Mass flow, t/h	**25.95**	**25.96**	**19.16**	**6.69**	**6.69**	**0.00**	**504.86**	**31.53**	**473.33**	**34.97**	**34.97**
Molar flow, kmol/h	824.3	824.4	447.3	371.4	371.4	0.0	21466.7	1548.5	20247.1	1940.9	1940.9
Temperature, °C	**101.1**	**41.1**	**41.1**	**41.1**	**41.1**	**35.0**	**116.1**	**120.0**	**120.0**	**138.9**	**138.9**
Pressure, kPa a	**186**	**175**	**175**	**175**	**325**	**400**	**188**	**188**	**188**	**350**	**350**
Pressure from	model	model	model	model	allowance	allowance	model	model	model	model	model
Density, kg/m³	1.881	3.883	2.870	991.8	991.8	994.1	1040.3	1.174	1023.8	1.841	926.1
Actual volume flow, m³/h	13797	6685	6678	6.75	6.75	0.00	485	26858	462	18997	38
Viscosity, mPa s	0.014	0.015	0.015	0.638	0.638	0.718	0.826	0.013	0.757	0.013	0.197
CO2, mol%	51.81	95.51	95.51					9.01		0.00	
H2O, mol%	48.19	4.49	4.49					90.99		100.00	
N2 + Ar, mol%	0.00	0.00	0.00					0.00		0.00	
O2, mol%	0.00	0.00	0.00					0.00		0.00	
Loading α, mol/mol							0.338		0.274		
MEA, wt% (apparent)							27.56		27.91		
CO2, wt% (apparent)							6.72		5.52		
H2O, wt% (apparent)							65.72		66.57		

Stream 11 is written as liquid as in the model: it flashes (EQ-P03). Stream 19 = 20 + 21 at steady state. Streams 3–5: assumed wash-water circulation. Pressure from: model = calculated by the model; allowance = the model's fixed allowance above column pressure; hydraulics = pump discharge (section 6).

Case C · Off-design 2 · 70 % turndown — streams 1 to 12

Stream no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
BFD no. (Process tab)		1				2	3	4		7	5	
Description	Flue gas from DCC	Flue gas to absorber	Wash water draw-off	Wash water to cooler	Cooled wash water	Treated gas to stack	Lean amine to absorber	Rich amine from sump	Rich amine pumped	Lean amine, warm	Rich amine, hot	Lean amine pumped
From	battery limit (DCC)	K-101	T-101 chimney tray	P-104	E-105	T-101	E-104 / FV-201	T-101	P-101 / FV-109	E-101	E-101	P-102
To	K-101	T-101	P-104	E-105	T-101 wash bed	stack	T-101	P-101	E-101	E-104	T-102	E-101
Phase	gas	gas	liquid	liquid	liquid	gas	liquid	liquid	liquid	liquid	liquid	liquid
Mass flow, t/h	**157.31**	**157.31**	**456.38**	**456.38**	**456.38**	**148.44**	**173.18**	**182.08**	**182.08**	**173.18**	**182.08**	**173.18**
Molar flow, kmol/h	5521.6	5521.6	—	—	—	5315.7	7441.4	7647.1	7647.1	7441.4	7647.1	7441.4
Temperature, °C	**40.0**	**40.0**	**45.1**	**45.1**	**37.0**	**40.0**	**40.0**	**43.1**	**43.1**	**49.6**	**110.0**	**120.0**
Pressure, kPa a	**102**	**105**	**103**	**469**	**418**	**103**	**403**	**120**	**487**	**453**	**295**	**469**
Pressure from	model	model	model	hydraulics	hydraulics	model	allowance	allowance	hydraulics	allowance	allowance	hydraulics
Density, kg/m³	1.119	1.150	990.2	990.2	993.4	1.105	1056.5	1109.3	1109.3	1051.2	1072.5	1012.5
Actual volume flow, m³/h	140523	136843	461	461	459	134364	164	164	164	165	170	171
Viscosity, mPa s	0.018	0.018	0.593	0.593	0.690	0.018	2.358	2.605	2.605	1.988	0.964	0.732
CO2, mol%	4.00	4.00				0.41						
H2O, mol%	7.07	7.07				7.21						
N2 + Ar, mol%	76.96	76.96				79.94						
O2, mol%	11.97	11.97				12.44						
Loading α, mol/mol							0.224	0.472	0.472	0.224	0.472	0.224
MEA, wt% (apparent)							28.23	26.87	26.87	28.23	26.87	28.23
CO2, wt% (apparent)							4.55	9.13	9.13	4.55	9.13	4.55
H2O, wt% (apparent)							67.22	64.00	64.00	67.22	64.00	67.22

Case C · Off-design 2 · 70 % turndown — streams 13 to LC

Stream no.	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**LS**	**LC**
BFD no. (Process tab)	8		10		9	12			6	11	
Description	Overhead vapour	Condenser outlet (two-phase)	CO2 product	Reflux water from drum	Reflux to stripper	Make-up water	Stripper bottoms to reboiler	Reboiler vapour	Lean amine from reboiler	LP steam	Steam condensate
From	T-102	E-103	V-101 / PV-301	V-101	P-103 / FV-303	demin header / FV-208	T-102 sump	E-102	E-102	LP steam header	E-102 (tubes)
To	E-103	V-101	compression	P-103	T-102	T-102 sump	E-102	T-102	P-102	E-102 (tubes)	condensate return
Phase	gas	two-phase	gas	liquid	liquid	liquid	liquid	gas	liquid	steam	liquid
Mass flow, t/h	**14.06**	**14.06**	**8.91**	**5.07**	**5.07**	**0.00**	**189.43**	**16.25**	**173.18**	**16.96**	**16.96**
Molar flow, kmol/h	493.7	493.7	207.7	281.5	281.5	0.0	8104.4	848.8	7441.4	941.5	941.5
Temperature, °C	**106.1**	**40.0**	**40.0**	**40.0**	**40.0**	**35.0**	**118.7**	**120.0**	**120.0**	**138.9**	**138.9**
Pressure, kPa a	**179**	**175**	**175**	**175**	**325**	**400**	**180**	**180**	**180**	**350**	**350**
Pressure from	model	model	model	model	allowance	allowance	model	model	model	model	model
Density, kg/m³	1.617	4.544	2.884	992.3	992.3	994.1	1023.6	1.055	1012.5	1.841	926.1
Actual volume flow, m³/h	8700	3095	3090	5.11	5.11	0.00	185	15399	171	9215	18
Viscosity, mPa s	0.014	0.015	0.015	0.652	0.652	0.718	0.766	0.013	0.732	0.013	0.197
CO2, mol%	40.29	95.76	95.76					4.35		0.00	
H2O, mol%	59.71	4.24	4.24					95.65		100.00	
N2 + Ar, mol%	0.00	0.00	0.00					0.00		0.00	
O2, mol%	0.00	0.00	0.00					0.00		0.00	
Loading α, mol/mol							0.270		0.224		
MEA, wt% (apparent)							27.97		28.23		
CO2, wt% (apparent)							5.43		4.55		
H2O, wt% (apparent)							66.60		67.22		

Stream 11 is written as liquid as in the model: it flashes (EQ-P03). Stream 19 = 20 + 21 at steady state. Streams 3–5: assumed wash-water circulation. Pressure from: model = calculated by the model; allowance = the model's fixed allowance above column pressure; hydraulics = pump discharge (section 6).

3 · Unit balances and duties

Envelope	In streams	Out streams	Case A in / out t/h	Case B in / out t/h	Case C in / out t/h	Error % A	Error % B	Error % C
T-101 absorber	2 + 7 + 5	6 + 8 + 3	950.91 / 950.91	1156.99 / 1157.57	786.86 / 786.89	0.000	−0.050	−0.004
T-102 + E-102	11 + 17 + 18	13 + 21	287.98 / 287.97	499.52 / 499.29	187.15 / 187.24	0.002	0.047	−0.047
E-102 kettle	19	20 + 21	291.36 / 291.36	504.86 / 504.86	189.43 / 189.43	0.000	0.000	0.000
T-102 column	11 + 17 + 18 + 20	13 + 19	309.63 / 309.63	531.04 / 530.81	203.40 / 203.49	0.002	0.044	−0.043
E-103 + V-101	13	15 + 16	18.27 / 18.27	25.95 / 25.85	14.06 / 13.98	−0.003	0.381	0.578
Overall unit	2 + 18	6 + 15	225.09 / 225.08	227.28 / 227.53	157.31 / 157.35	0.002	−0.109	−0.022

CO2 balance	In t/h	Stack t/h	Product t/h	Closure %	Capture %
Case A	13.886	1.385	12.501	0.000	90.03
Case B	20.829	2.062	18.801	−0.165	90.10
Case C	9.720	0.970	8.754	−0.038	90.02

Item	Unit	Case A	Case B	Case C	Governing
E-101 lean/rich exchanger	MW	18.45	30.09	12.19	B
E-102 reboiler	MW	13.69	20.86	10.12	B
E-103 overhead condenser	MW	4.08	5.03	3.76	B
E-104 lean cooler	MW	3.13	6.65	1.66	B
E-105 wash-water cooler	MW	5.77	8.24	4.28	B
K-101 blower shaft power	kW	313	334	131	B
CW to E-103: flow kg/s / rise K		221 / 4.4	400 / 3.0	160 / 5.6	
CW to E-104: flow kg/s / rise K		509 / 1.5	1000 / 1.6	193 / 2.1	
CW to E-105: flow kg/s / rise K		249 / 5.5	302 / 6.5	255 / 4.0	

4 · Off-design cases: what changes

Result	Case A · Design case	Case B · Off-design 1 · 6 % CO2, lean raised	Case C · Off-design 2 · 70 % turndown
Lean flow FIC-201, t/h	269.7	473.3	173.2
Rich flow FI-109, t/h	282.2	492.8	182.1
Lean / rich loading	0.246 / 0.471	0.274 / 0.471	0.224 / 0.472
Steam FIC-207, t/h	22.94	34.97	16.96
Reboiler duty, MW	13.69	20.86	10.12
SRD, GJ/t	3.94	3.99	4.16
CO2 product, t/h	12.50	18.80	8.75
Temperature bulge, °C	57.3	64.1	56.7
Rich to T-102 TI-204, °C	109.0	107.7	110.0
T-101 / T-102 flood, %	62.1 / 50.6	68.2 / 76.7	43.4 / 36.6
FV-201 / FV-207 / FV-109 opening, %	52 / 49 / 52	91 / 74 / 90	33 / 36 / 33
TV-202 / TV-302 / TV-110 opening, %	51 / 55 / 50	100 / 100 / 60	19 / 40 / 51
Lean to T-101, °C	40.0	42.4	40.0
Blower speed / power	80 % / 313 kW	81 % / 334 kW	58 % / 131 kW

Case B (6 % CO2) needs 1.76 × the design lean flow and governs the pumps, the amine lines, E-101, E-102 and the coolers; it saturates the CW valves TV-202 and TV-302 (EQ-P05). Case C (70 %) runs the absorber at 43 % of flood and FV-201 at a third open: the minimum-flow and wetting checks of the packing belong to the Design tab.

5 · Line list and line sizing

Sizing flow = the largest actual volume flow of the three cases × 1.10; the smallest standard size meeting every typical criterion is selected (SS Sch 10S, CS STD; ducts as round equivalents). Line numbers follow CC-PID-101 (DNxxx-SVC-nnn, sizes TBC); "new" numbers are given here for lines the P&ID does not number.

Line	Service	Strm	Material	Gov. case	Sizing t/h	m³/h	ρ kg/m³	Criterion (v max m/s · ΔP max kPa/100 m)	Selected	v m/s	ΔP kPa/100 m	v design	P&ID DN (TBC)	v / ΔP at drawn DN
DUCT-FG-101 (new)	Flue gas K-101 → T-101	2	CS + lining (TBC)	A	247.32	211125	1.17	Flue-gas duct (20 · —)	**Ø 2000 mm**	18.67	0.11	16.97	—	—
DUCT-FG-109 (new)	Treated gas T-101 → stack (new no.)	6	SS 316L	A	233.59	211446	1.10	Flue-gas duct (20 · —)	**Ø 2000 mm**	18.70	0.11	17.00	—	—
RA-103 (new)	Rich amine T-101 sump → P-101 (new no.)	8	SS 316L	B	542.11	490	1107	Pump suction (1.0 · 10)	**DN450**	0.87	1.55	0.45	—	—
RA-102	Rich amine P-101 → FV-109 → E-101	9	SS 316L	B	542.11	490	1107	Pump discharge (2.5 · 40)	**DN250**	2.47	20.55	1.28	DN400	1.10 / 2.79
RA-104	Hot rich amine E-101 → T-102	11	SS 316L	B	542.11	505	1073	Hot rich amine (1.5 · 20)	**DN350**	1.49	5.05	0.77	DN400	1.13 / 2.55
LA-202 (new)	Lean amine E-102 → P-102 (new no.)	21	SS 316L	B	520.67	509	1024	Suction, saturated (0.8 · 5)	**DN500**	0.73	0.79	0.38	—	—
LA-201	Lean amine P-102 → E-101	12	SS 316L	B	520.67	509	1024	Pump discharge (2.5 · 40)	**DN300**	1.82	7.70	0.94	DN400	1.14 / 2.41
LA-203	Lean amine E-101 → E-104	10	SS 316L	B	520.67	492	1059	Pump discharge (2.5 · 40)	**DN250**	2.48	19.27	1.29	DN400	1.10 / 2.59
LA-204	Lean amine E-104 → FV-201 → T-101	7	SS 316L	B	520.67	488	1067	Pump discharge (2.5 · 40)	**DN250**	2.46	19.66	1.28	DN400	1.10 / 2.67
LA-205 (new)	T-102 sump → E-102 (new no.)	19	SS 316L	B	555.34	534	1040	Gravity liquid (1.0 · 5)	**DN450**	0.94	1.49	0.50	—	—
VR-302	Reboiler vapour E-102 → T-102	20	SS 316L	B	34.68	29544	1.17	LP vapour (25 · 3)	**DN700**	21.62	0.50	14.52	DN500	42.32 / 2.70 ✗
OV-301	Overhead vapour T-102 → E-103	13	SS 316L	B	28.55	15176	1.88	LP vapour (25 · 3)	**DN500**	21.74	1.16	14.97	DN500	21.74 / 1.16
OV-305 (new)	Condenser outlet E-103 → V-101 (new no.)	14	SS 316L	B	28.55	7353	3.88	Two-phase (15 · 5)	**DN450**	12.99	0.96	7.81	—	—
CO-304	CO2 product V-101 → PV-301 → compression	15	SS 316L	B	21.08	7346	2.87	Wet CO2 gas (20 · 5)	**DN400**	16.50	1.33	9.91	DN300	26.22 / 4.30 ✗
RF-306 (new)	Reflux water V-101 → P-103 (new no.)	16	SS 316L	B	7.36	7.42	992	Suction, saturated (0.8 · 5)	**DN80**	0.38	2.00	0.29	—	—
RF-303	Reflux P-103 → FV-303 → T-102	17	SS 316L	B	7.36	7.42	992	Pump discharge (2.5 · 40)	**DN50**	0.88	15.56	0.66	DN50	0.88 / 15.56
DW-208	Make-up water → FV-208 → T-102	18	SS 316L	A	0.27	0.27	994	Pump discharge (2.5 · 40)	**DN25**	0.12	1.11	0.11	DN25	0.12 / 1.11
WW-106 (new)	Wash water T-101 draw-off → P-104 (new no.)	3	SS 316L	B	502.01	509	987	Pump suction (1.0 · 10)	**DN450**	0.90	1.23	0.82	—	—
WW-105	Wash water P-104 → E-105 → T-101	4	SS 316L	B	502.01	509	987	Pump discharge (2.5 · 40)	**DN300**	1.82	7.25	1.65	DN200	4.02 / 54.93 ✗
LS-207	LP steam header → FV-207 → E-102	LS	CS	B	38.46	20897	1.84	LP steam (35 · 10)	**DN500**	30.92	2.28	18.44	DN400	49.26 / 7.48 ✗

Criteria are typical practice, not a company standard. Darcy friction factor by Swamee–Jain, roughness 0.045 mm. ✗ = the drawn (TBC) size does not meet the criterion at the sizing flow (EQ-P01).

6 · Pump and blower hydraulics

Head = (destination − source surface pressure) + static head + line friction + equipment ΔP + control-valve ΔP; NPSHa = $(P_{surface} - P_{vap})/\rho g + (z_{LLL} - z_{pump}) - h_{f,suction}$ at rated flow. Elevations and line lengths are assumed (below); equipment ΔP are datasheet allowances at the rated flow; control valve = one third of the total dynamic loss, at least 70 kPa (typical).

Pump (case A hydraulics)	**P-101**	**P-102**	**P-103**	**P-104**
Service	Rich amine pump	Lean amine pump	Reflux pump	Wash water pump
Source → destination	T-101 sump → T-102 rich feed	E-102 lean compartment (saturated) → T-101 lean distributor	V-101 (saturated with CO2) → T-102 reflux distributor	T-101 wash draw-off → T-101 wash distributor
Liquid T °C / ρ kg/m³	43.6 / 1112	120.0 / 1020	40.0 / 992	47.9 / 989
Normal / rated flow, m³/h	253.7 / 487.5	264.4 / 510.4	5.6 / 7.4	461.4 / 507.6
Normal / rated flow, t/h	282.2 / 542.1	269.7 / 520.7	5.5 / 7.4	456.4 / 502.0
Source P / vapour P, kPa a	107.0 / 11.6	182.4 / 182.4	175.0 / 175.0	103.6 / 11.2
Destination P, kPa a	180.9	103.6	180.9	103.0
Static ΔP, kPa	251.1	139.3	178.1	33.3
Suction friction N / R, kPa	0.3 / 1.1	0.2 / 0.6	0.2 / 0.3	1.0 / 1.2
Discharge friction N / R, kPa	3.4 / 11.5	6.4 / 22.7	3.7 / 6.2	4.8 / 5.8
Equipment + FE N / R, kPa	21.7 / 80.0	34.9 / 130.0	5.7 / 10.0	41.3 / 50.0
Control valve allowance, kPa	70	76	70	none
Differential pressure N / R, kPa	347 / **414**	257 / **369**	258 / **265**	81 / **90**
Head N / R, m	31.8 / **37.9**	25.7 / **36.9**	26.5 / **27.2**	8.3 / **9.3**
Discharge pressure (normal), kPa a	504	495	473	468
Efficiency (typical) / absorbed kW	0.78 / 71.8	0.78 / 67.1	0.35 / 1.6	0.78 / 16.3
Motor, kW	**90**	**90**	**2.2**	**22**
NPSHa / NPSHr (assumed), m	**12.16** / 4.0 ✓	**5.24** / 4.0 ✓	**3.69** / 2.5 ✓	**38.50** / 3.0 ✓
Lowest LLL for NPSHr + 1 m / used, m	−3.04 / 4.12	5.66 / 5.90	4.13 / 4.32	−4.80 / 29.70
Shut-off ΔP / max discharge P, kPa a	497 / **677**	443 / **868**	318 / **713**	108 / **512**

Pressure profiles along the discharge path (normal flow, case A)

P-101: point	z m	P kPa a
Pump discharge	0.6	504
end of RA-102 (incl. FE-109 orifice)	1.0	496
FV-109 outlet (ΔP 70 kPa)	1.0	426
end of RA-102	1.5	419
E-101 rich side outlet	1.5	400
T-102 rich feed	21.5	181
HMB stream 11 (E-101 rich outlet): model allowance / hydraulics		295 / 400

P-102: point	z m	P kPa a
Pump discharge	0.6	495
end of LA-201	1.5	486
E-101 lean side outlet	1.5	467
end of LA-203	1.5	465
E-104 tube side outlet	1.5	452
end of LA-204 (incl. FE-201 orifice)	1.0	453
FV-201 outlet (ΔP 76 kPa)	1.0	376
T-101 lean distributor	28.0	104
HMB stream 10 (E-104 inlet): model allowance / hydraulics		453 / 465
HMB stream 7 (FV-201 inlet): model allowance / hydraulics		403 / 453

P-103: point	z m	P kPa a
Pump discharge	0.6	473
end of RF-303 (incl. FE-303 orifice)	1.0	463
FV-303 outlet (ΔP 70 kPa)	1.0	393
T-102 reflux distributor	22.5	181
HMB stream 17 (FV-303 inlet): model allowance / hydraulics		325 / 463

P-104: point	z m	P kPa a
Pump discharge	0.6	468
end of WW-105	1.5	459
E-105 process side outlet	1.5	417
T-101 wash distributor	33.5	103

Elevation (assumed), m above grade	m
T-101 bottom tangent / LLL / NLL / HLL	4 / 4.12 / 5.25 / 6.25
T-101 lean inlet / wash draw-off / wash inlet / top tangent	28.0 / 30.0 / 33.5 / 35.0
T-102 bottom tangent / rich feed / reflux / top tangent	9.5 / 21.5 / 22.5 / 24.0
E-102 shell bottom / lean LLL	5.5 / 5.90
V-101 bottom / LLL / NLL	4 / 4.32 / 4.80
Pumps / exchangers / control valves	0.6 / 1.5 / 1

Blower K-101 (model curve)	
Suction flow, m³/h (40 °C)	200747
Pressure rise, kPa	4.70
Speed / shaft power	80 % / 313 kW (η 0.80 in the model)
Shut-off rise at 100 % speed H_0 → T-101 max	16.2 kPa → 118.5 kPa a
100 % speed from the design state (model run)	126 % of design flow, 589 kW

7 · Design margins

Item	Unit	Normal (A)	Case B	Case C	Rated	Capacity / limit	Margin on governing	Basis
Lean amine P-102 / FV-201	flow t/h	269.7	473.3	173.2	520.7	520.0	9.9 % ✗	rated = 1.10 × governing case; capacity = FV-201 wide open (K.FmaxL, model, assumed)
Rich amine P-101 / FV-109	flow t/h	282.2	492.8	182.1	542.1	545.0	10.6 %	capacity = FV-109 wide open (K.FmaxR)
Reflux P-103 / FV-303	flow t/h	5.5	6.7	5.1	7.4	10.8	61.4 %	capacity = FV-303 wide open (K.FmaxRf)
LP steam FV-207	flow t/h	22.9	35.0	17.0	38.5	47.0	34.4 %	capacity = FV-207 wide open at 3.5 bar a (K.FmaxS)
Reboiler E-102 duty	MW	13.7	20.9	10.1	22.9	26.5	26.8 %	capacity = UA·(T_steam − T_reb) at design (K.UAreb 1400 kW/K, model)
Flue gas K-101	kNm³/h	176.8	176.8	123.8	—	223.1	26.2 %	capacity = blower at 100 % speed from the design state (model run)
T-101 flooding AY-102	% flood	62.1	68.2	43.4	—	80.0	11.8 %	limit = ALM AY-102 HI 80 % (typical design 70–80 %)
T-102 flooding AY-210	% flood	50.6	76.7	36.6	—	80.0	3.3 %	limit = ALM AY-210 HI 80 %
E-104 lean cooler TV-202	% open	50.9	100.0	19.3	—	95.0	−5.0 % ✗	soft limit: TIC-202 loses its set point above ≈ 95 %
E-103 condenser TV-302	% open	55.2	100.0	40.0	—	95.0	−5.0 % ✗	soft limit: TIC-302 loses its set point
Lean to T-101 TIC-202	°C	40.0	42.4	40.0	—	45.0	2.6 °C	ALM TIC-202 HI 45 °C

Rated = 1.10 × the governing case. Capacity = what the installed item gives in the model. ✗ = the design basis and the equipment do not agree (EQ-P04, EQ-P05).

8 · Assumptions and holds (not design data)

1. Wash-water circulation (not in the model): set at design from approach temperatures — supply = TIC-110 − 3 K, return = top-of-bed gas − 5 K → 456 t/h; kept constant in the off-design cases.

2. Elevations above grade (T-101 bottom tangent 4.0 m, T-102 bottom tangent 9.5 m, E-102 shell bottom 5.5 m, V-101 bottom 4.0 m, pumps 0.6 m, exchangers 1.5 m, control valves 1.0 m) and equivalent line lengths.

3. Equipment ΔP allowances at rated flow: E-101 70 kPa per side (plate), E-104 50 kPa, E-105 50 kPa, flow elements 10 kPa; control valve = one third of the total dynamic loss, at least 70 kPa (typical).

4. Pump rated flow = 1.10 × the governing case; shut-off ΔP = 1.2 × rated ΔP; efficiency by size (0.35–0.78) and NPSHr (2.5–4 m) typical, not vendor data.

5. Line sizing flow = 1.10 × the governing case; roughness 0.045 mm; SS lines Sch 10S, CS lines STD; flue-gas and treated-gas ducts as round equivalents.

6. Gas viscosities 0.013–0.018 mPa s (typical); water properties from correlations; amine density/viscosity from the model (rhoL, muL).

7. Liquid pressures of streams 7, 8, 10, 11, 17, 18 are the model's fixed allowances (Process tab, PFD flags); streams 4, 5, 9 and 12 are pump discharge pressures from part 5.

8. LP steam 3.5 bar a saturated and CW 30 °C (assumed, TBC); PSV-301 set 350 kPa a (assumed, HOLD on CC-PID-101) used as the T-102 system maximum for shut-off pressures.

Holds: PSV set pressures (CC-PID-101 note 5), the solvent filters, reclaimer and storage (note 7), measured VLE at the stripper inlet, the CW and LP-steam conditions.

9 · Engineering queries

EQ	What	Documents	Finding	Suggested resolution
EQ-P01	Line sizes on CC-PID-101 (all marked TBC) disagree with the sizing in part 4	CC-PID-101 rev A vs MEA-PE-HMB-001	Amine lines drawn DN400 need DN250–300; VR-302 DN500 gives 42 m/s (needs DN700); LS-207 DN400 → DN500; WW-105 DN200 → DN300; CO-304 DN300 → DN400.	Issue the line list with the sizes of part 4 and update the P&ID at rev B.
EQ-P02	Liquid pressures in the HMB/PFD (model allowances) are below the pump hydraulics	MEA-PFD-001 flags, Process tab vs part 5	Stream 11 (E-101 rich outlet) 295 vs 400 kPa a; stream 17 (reflux) 325 vs 463 kPa a; stream 7 (FV-201 inlet) 403 vs 453 kPa a.	Keep the model allowances for the OTS; mark them "allowance" on the PFD and take pump-side pressures from MEA-PE-HMB-001.
EQ-P03	Hot rich amine flashes after E-101	Model VLE vs CC-PID-101 (FV-109 upstream of E-101)	Model bubble pressure of the rich solvent at 109 °C, α 0.471 is 1352 kPa a, above the line pressure (400 kPa a): the rich side of E-101 and RA-104 run two-phase. The correlation is extrapolated (TBC) but flashing here is normal practice.	Move the rich letdown (FV-109) to the T-102 inlet or add a back-pressure valve there; size RA-104 for two-phase flow; confirm with measured VLE.
EQ-P04	The 6 % CO2 case sets the pumps and P-102 has no margin on FV-201	Part 6 vs model valve capacity	Lean flow at 6 % CO2 is 473 t/h (1.76 × design); rated P-102 521 t/h equals FV-201 wide open (K.FmaxL 520 t/h). At design P-102 runs at 52 % of rated flow.	Confirm the 6 % CO2 design basis; consider 2 × 60 % pumps and a larger FV-201 trim; check the minimum continuous flow.
EQ-P05	Cooling water saturates at 6 % CO2	Part 6 (case B)	TV-202 and TV-302 reach 100 % and 100 %; lean to T-101 42.4 °C (TIC-202 SP 40). The CW flows in the model are assumed (1000/400/500 kg/s) and give a CW rise of only 1.5 K in E-104 at design.	Set CW flows for a typical 8–10 K rise and size E-103/E-104 for case B; update the model constants at the next model revision.
EQ-P06	P-102 NPSH needs an elevated kettle	Part 5 vs layout	NPSHa = static head only (saturated lean). E-102 low level must be at ≥ 5.66 m above grade (5.90 m used) and the T-102 bottom tangent above the kettle liquid level.	Fix E-102 at +5.5 m and T-102 at +9.5 m in the Layout tab and the GA (Mechanical tab).

The tab "Process engineering" of the OTS recomputes every check of this document in the page (stream table against the model state, PFD flags, unit balances, line sizing, pump hydraulics, margins).